

of HT12D decoder on the receiver side. The two data pins of the HT12D are connected to two interrupt pins of the Arduino UNO board.

When a visitor passes through the entry gate, a pulse is generated at pin 7 (OUT2) of LM358, which is connected to AD9 pin of HT12E. The same pulse

is transmitted and appears on D9 pin of HT12D on the receiver. The D9 pin is connected to interrupt pin 3 of the Arduino board. The interrupt subroutine executes and increments the visitor count by one. The operation is same for the exit of a visitor except that separate data lines of HT12E (pin AD8) and HT12D (pin D8) are used and pin 2 of the Arduino is used as the interrupt pin.

The program running in the microcontroller of the Arduino monitors the interrupts and increments/decrements the visitors count and displays it on the 16 × 2 LCD. Arduino IDE is used for programming the Arduino Uno board.

Construction and testing

Connect the Arduino board to PC and select proper COM port and board

Expected Signals at Test Points

Table 1

Test point	Expected signal
TP1	9V
TP2	GND (0V)
TP3	5V
TP4	Train of pulses
TP5	Low-high-pulse when a visitor enters
TP6	Low-high-pulse when a visitor leaves
TP7	Low-high-pulse when a visitor leaves
TP8	Low-high-pulse when a visitor enters
TP9	9V
TP10	GND (0V)
TP11	5V
TP12	Train of pulses
TP13	Low-high-pulse when a visitor enters
TP14	Low-high-pulse when a visitor leaves

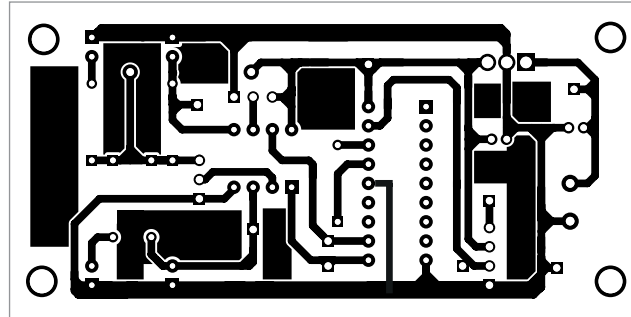


Fig. 7: Actual-size PCB layout for transmitter

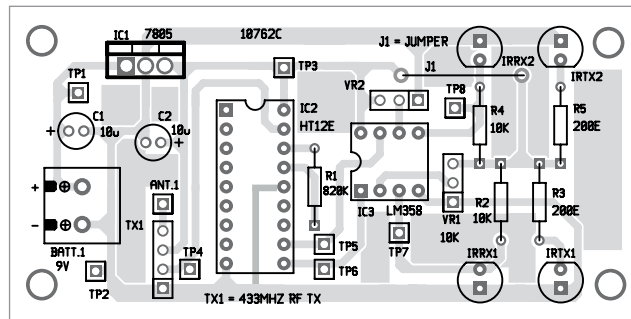


Fig. 8: Components layout of the transmitter PCB

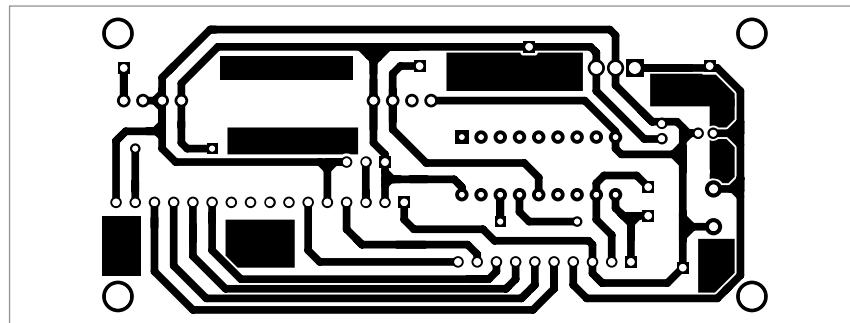


Fig. 9: Actual-size PCB layout for receiver

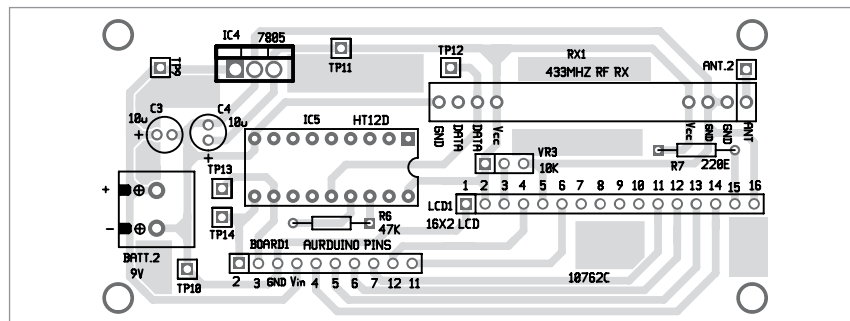


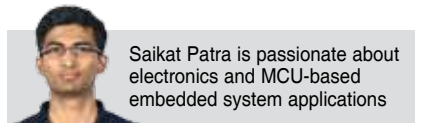
Fig. 10: Components layout of the receiver PCB

name from Tools menu of the Arduino IDE. Then upload the source code *visitor_counter_wl.ino* to the Arduino board by clicking on Upload button of the IDE.

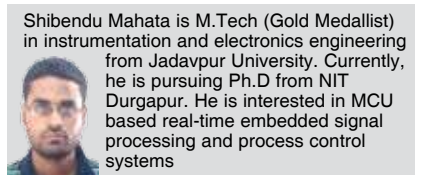
An actual-size PCB layout for the transmitter is shown in Fig. 7 and its components layout in Fig. 8. After assembling the circuit on PCB, connect a 9V battery across BATT.1.

An actual-size PCB layout for the receiver is shown in Fig. 9 and its components layout in Fig. 10. After assembling the circuit on PCB, connect a 9V battery across BATT. 2.

For troubleshooting, check the signals at different test points marked in Fig. 5 and Fig. 6 using a multimeter. The expected signals at different test points are listed in the Table. **EFY**



Saikat Patra is passionate about electronics and MCU-based embedded system applications



Shibendu Mahata is M.Tech (Gold Medallist) in instrumentation and electronics engineering from Jadavpur University. Currently, he is pursuing Ph.D from NIT Durgapur. He is interested in MCU based real-time embedded signal processing and process control systems